

Chapter 1

Introduction to Networking

What is Computer Network?



- Computer network is an interconnection of two or more computers and peripherals such as printers and faxes.
- Computer network allows the user to share and transfer information using cables or modems within a network.

Communication

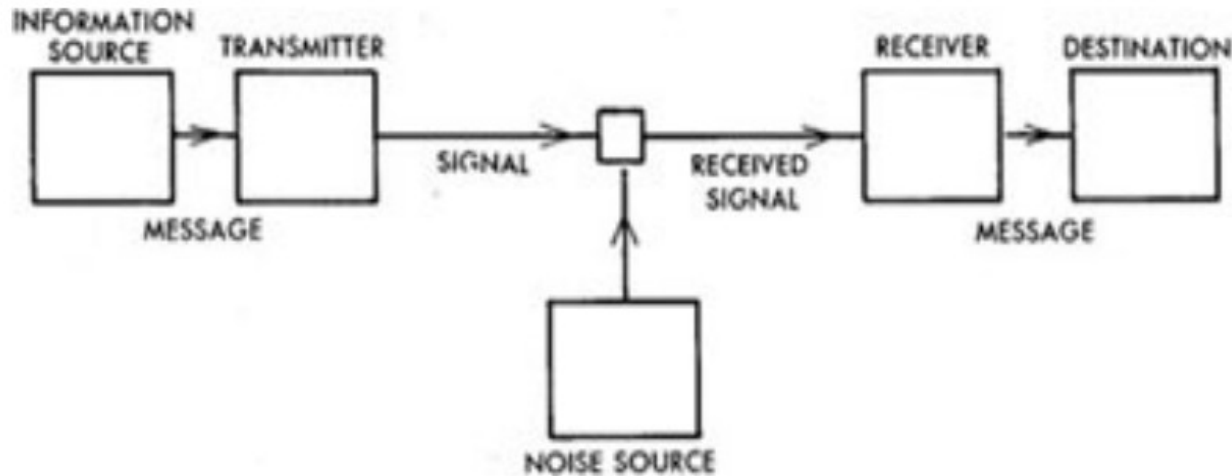
- Exchange of information through the use of speech, signs or symbols is called communication.
When early humans started speaking, some 5,00,000 years ago, that was the first mode of communication.
- Before we dive into modern technologies that drive communication in contemporary world, we need to know how humans developed better communication techniques to share knowledge with each other.

History of Communication

- Communicating with people over a distance is known as **telecommunication**. The first forms of telecommunication were **smoke signals**, **drums** or fire **torches**.
- The major disadvantage with these communication systems was that only a set of pre-determined messages could be transmitted. This was overcome in the 18th and 19th century through development of **telegraphy** and **Morse code**.

- Invention of telephone and establishment of commercial telephony in 1878 marked a turnaround in communication systems and real telecommunication was born.
- International Telecommunication Union (ITU) defines telecommunication as transmission, emission and reception of any signs, signals or messages by electromagnetic systems. Now we had the

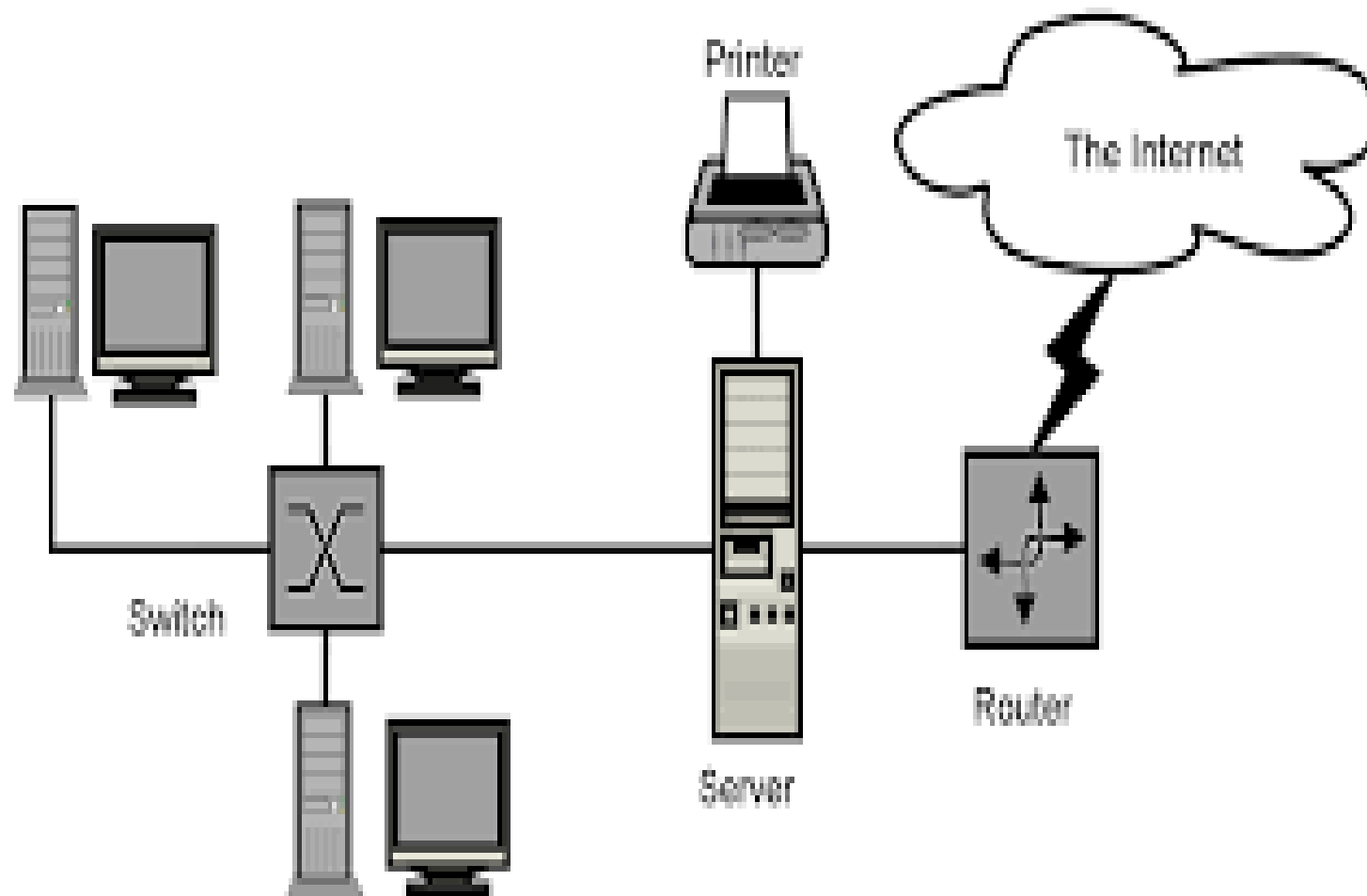
Communication components



- Message /data
- Source
- Medium
- Destination
- Protocol

Physical components of a network

- **Pc**
- **Interconnections** —components that provide means for data to travel from one point to another point over the local network.
 - Network interface card(NIC)
 - Network Media
 - Cables
 - Wireless media
 - Connector —provide connection point for media
- **Switches** —device that provide network attachment to the end systems and intelligent switching for data with in local network
- **Routers** —for internetworking and path selection



Why Networking ?

- Allows File Sharing
- Provides Convenient Resource Sharing(camera, printer, network storage, backup devices)
- Inexpensive System
- Flexible to be Used
- Increase in Storage Capacity of the Software

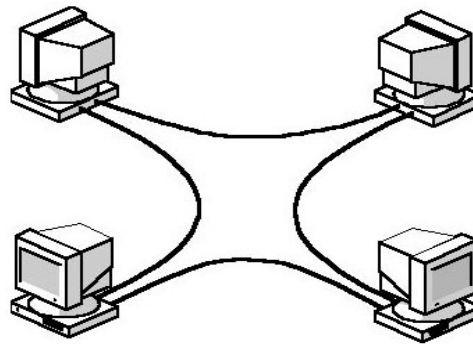
Introduction to Computer Network

Disadvantages

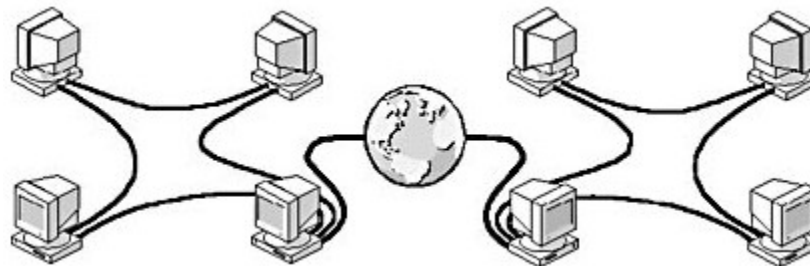
- **Security Difficulties**
- **Presence of Computer Viruses and Other Malwares**
- **Lack of Independence**

Network Categories:

- based on size, ownership, the distance it covers
 - **Local Area Network (LAN):** usually privately owned and links devices in a single office, building or campus



- **Wide Area Network (WAN):** covering large geographic area; may utilize public, leased, or private communications equipment

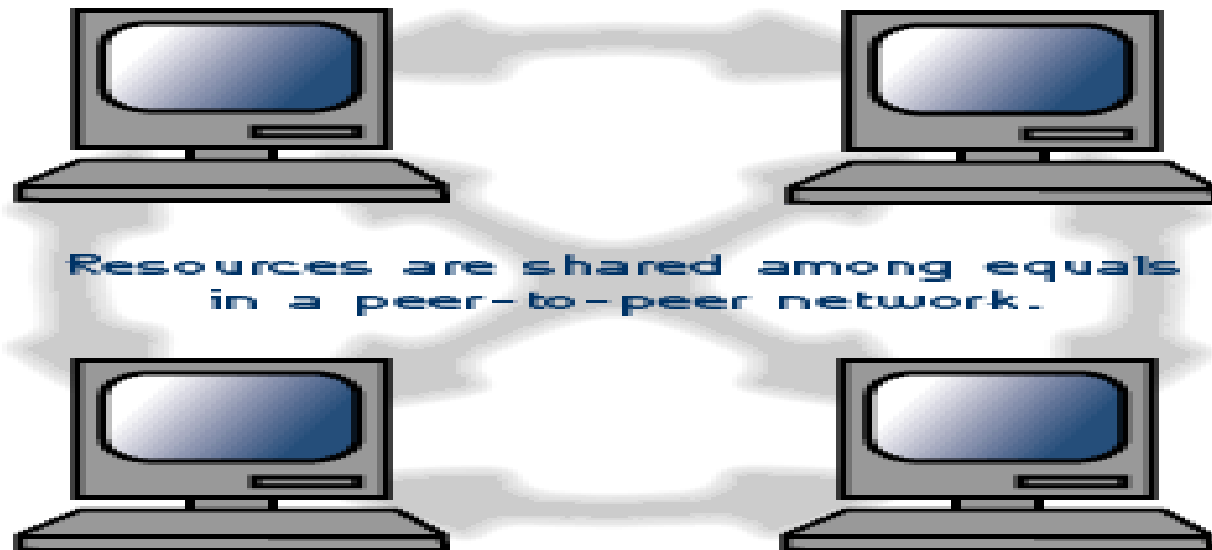


- **Metropolitan Area Network (MAN):** designed to extend over an entire city; it may be a single network or interconnected LANs
- **Personal Area Network (PAN):** meant for one person; e.g. a wireless network connecting a computer with its mouse, keyboard and printer

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

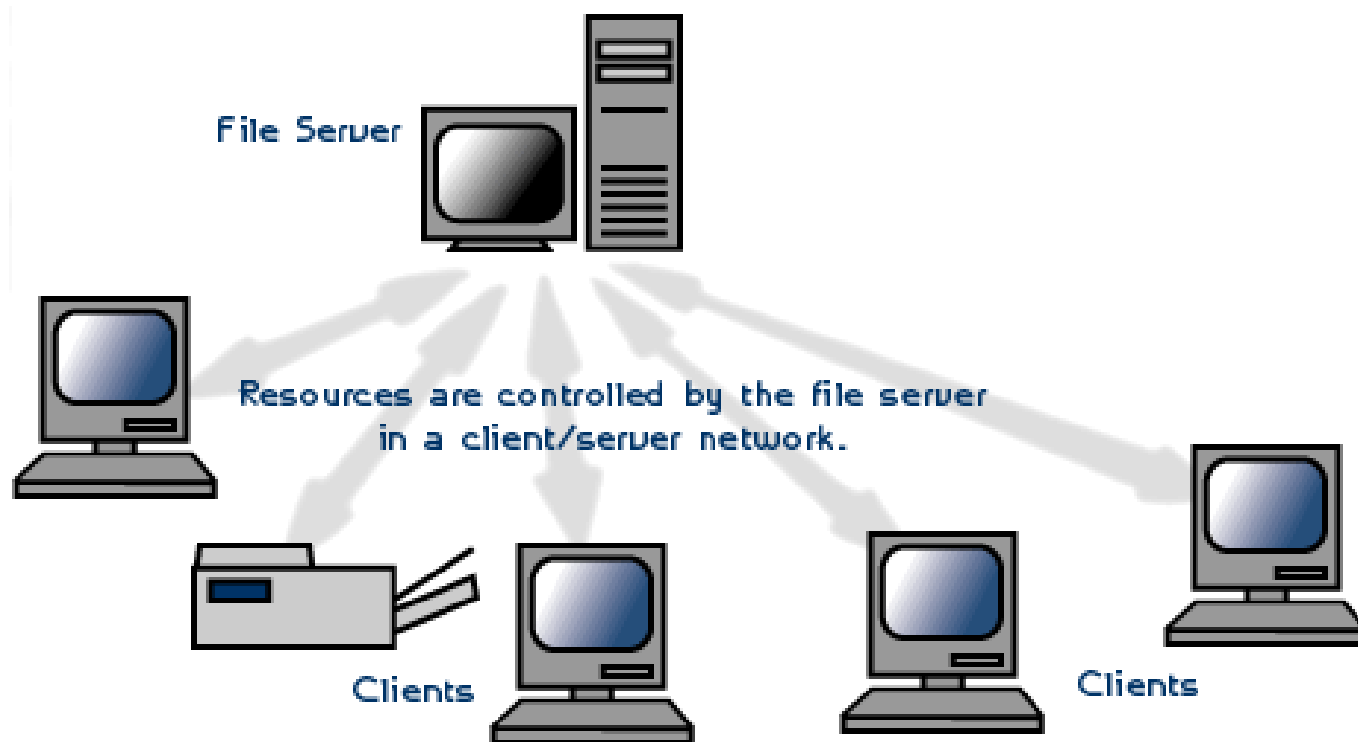
- **Network Architecture** : includes the type of computers on the network and determines how network resources are handled
- Two common types
 - Peer-to-peer
 - Client/Server
- **Peer-to-peer**
 - Each node considered as equal in terms of resource sharing and responsibilities
 - **pros**
 - Easy to set up
 - Less expensive
 - Demands moderate level of skill to administer
 - User is able to control their own resources
 - **Cons**
 - Only < 10 nodes
 - Very low level of security
 - Performance suffers when a computer is accessed

- Peer-to-peer networks are good choices for environments where:
 - There are 10 users or fewer
 - Users share resources, such as printers, but no specialised servers exist
 - Security is not an issue
 - The organization and the network will experience only limited growth within the foreseeable future



- **Client Server Model**

- Consists of a group clients connected to a server
- Server – with more RAM, larger hard disk, more processing power...



servers in networking

1. File and Print Servers

- File and print servers manage user access and use of file and printer resources.

2. Application Servers

- Application servers make the server side of client/server applications, as well as the data, available to clients.
 - an application server differs from a file and print server. With a file and print server, the data or file is downloaded to the computer making the request. With an application server, the database stays on the server and only the results of a request are downloaded to the computer making the request.

3. Mail Servers

- Mail servers operate like application servers in that there are separate server and client applications, with data selectively downloaded from the server to the client.

4. Fax Servers

- Fax servers manage fax traffic into and out of the network by sharing one or more fax modem boards.

5. Directory Services Servers

- Directory services servers enable users to locate, store, and secure information on the network.
- **Advantages of client/server architecture**
 - Sharing Resources
 - Security
 - Number of Users - A server-based network can support thousands of users
- **Cons**
 - more complex to install, configure, and manage
 - Expensive

Transmission media

- Is a physical media that carries a signal from the transmitter to the receiver
- The information or signal transmitted from one device to another is through electromagnetic signals.
- Electromagnetic signals include power, voice, radio, waves, infrared light, visible light, ultraviolet light, X-rays, and gamma rays.
- These signals can travel through vacuum, air or any other transmission medium
- The measurement of the quantity of data that can be passed down (transmitted) a communication link in a given time is done in terms of *bandwidth*
- In digital circuits, bandwidth is measured in *bits per second* (bps)

- Two basic categories
 - Guided
 - Unguided
 - **Guided** – uses a cabling system that guides the signals along a specific path
 - E.g. Fiber Optics, Twisted Pair, Coaxial Cable etc...
 - **Unguided** – consists of a means for the data signals to travel but nothing to guide them along a specific path - wireless
 - Example: Radio wave, Satellite, etc.

- **Twisted Pair**
 - Consists of two insulated copper wires twisted around each other
 - Twisting cancels out electrical noise from adjacent pairs (crosstalk) and external sources
 - Uses RJ-45 telephone-type connectors (larger than telephone and consists of eight wires Vs. Telephone's 4 wires)
 - Generally inexpensive
 - Easy to install
 - Up to 100 meters without a repeater
 - Can be either Shielded or Unshielded
 - STP (Shielded Twisted Pair)
 - UTP (Unshielded Twisted Pair)



Cable

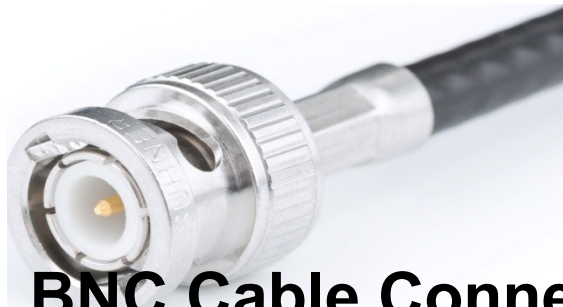
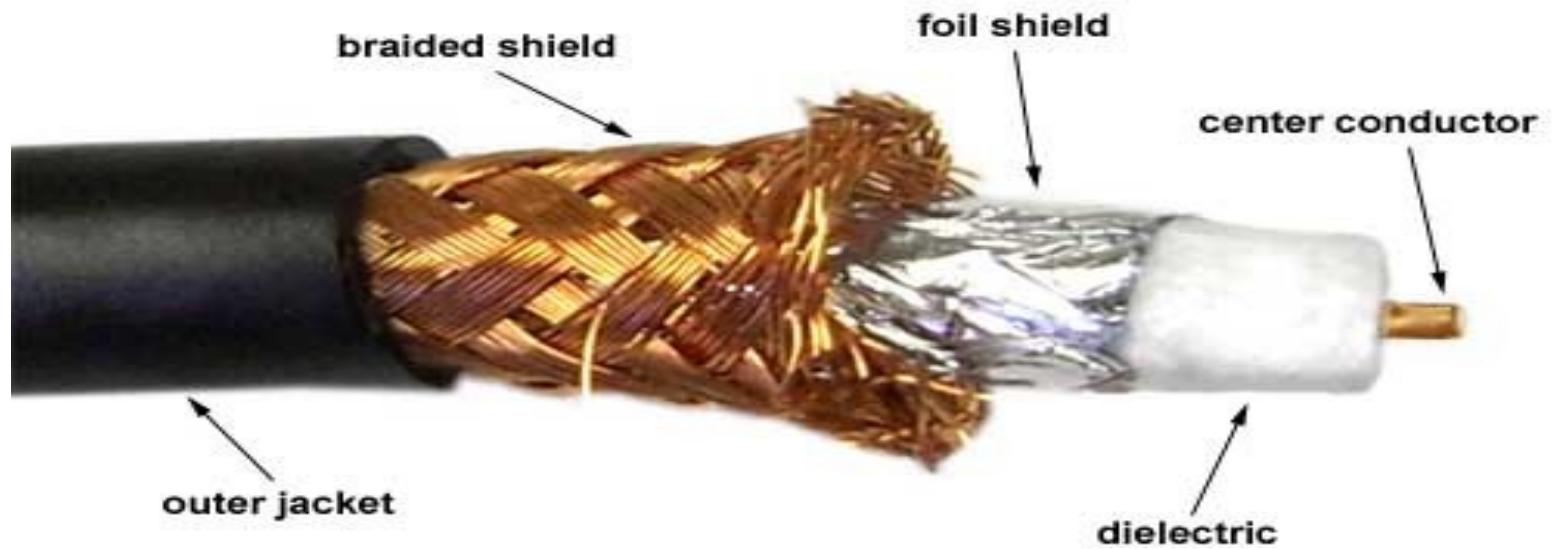
RJ-45 Connector



- **Coaxial Cable**

- Consists of a solid or stranded copper core surrounded by insulation, a braided shield and an insulating jacket
- Braided shield prevents noise and crosstalk
- More resistant to interference and attenuation than twisted pair cabling
- Can use
 - BNC cable connector
 - BNC Barrel connectors
 - BNC T connectors
 - BNC Terminators
- Can transmit data, voice and video
- Up to 200/500 meters without a repeater depending on the type of cable

COAXIAL CABLE



BNC Cable Connector



BNC Barrel Connector



BNC T-Connector

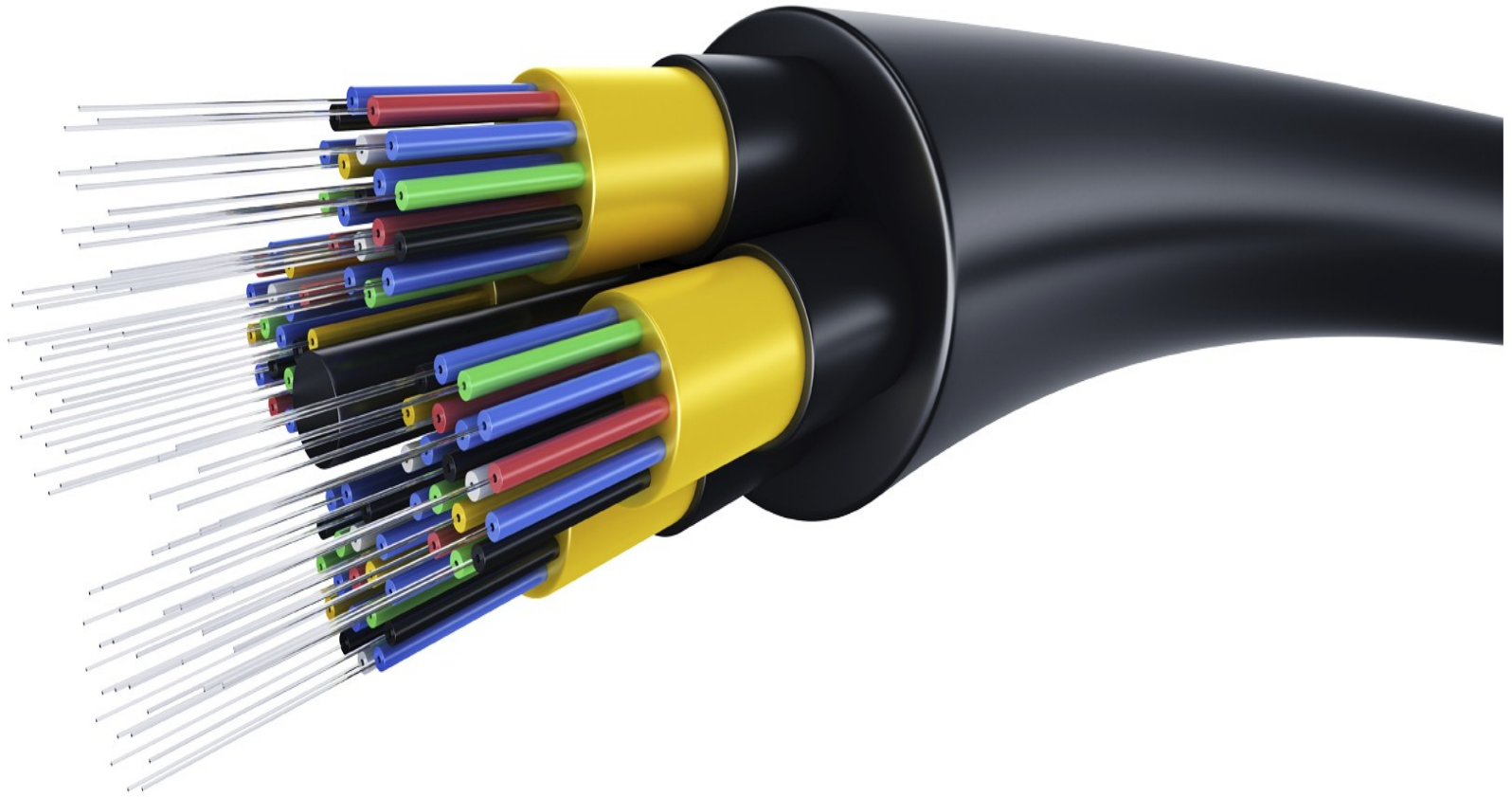


BNC Terminator

- **Fiber Optic Cable**

- Consists of a small core of glass or plastic surrounded by a cladding layer and jacket
- Fibers are unidirectional (light only travels in one direction) so two fibers are used, one for sending and one for receiving
- Good for very high speed, long distance data transmission
- Not subjected to electrical interference
- Cable cannot be tapped and data stolen => high security
- Up to 2000 meters without a repeater
- Most expensive and difficult to work with.

Fiber Optics Cable



Fiber Optics Cable Connectors



ST



SC



FC



LC



MPO



ESCON



MTRJ



VF45